ActInf Livestream #018.0 ~ The predictive global neuronal workspace: A fo

Livestream | The predictive global neuronal workspace: A formal act inf model | 1:24:06

hello and welcome
i think that we're going now this is the
active inference live stream
it's active inference live stream 18 on
march
17 2021 welcome to the active inference
lab everyone
we are a participatory online lab that
is communicating
learning and practicing applied active
inference you can find us at any of
these links
here this is a recorded and an archived
live stream
so please provide us with feedback so
that we can improve on our work
all backgrounds and perspectives are
welcome here and we'll be using good
video etiquette for live streams and
stephen thanks so much for
joining always awesome to have you on
for the conversations
and uh this will be very fun convo
here looking at the calendar we can see
at the short and linked that we're
down in the end of march 2021 in
number 18 which is going to be
about the paper the predictive global
neuronal workspace
a formal active inference model of
visual consciousness
by christopher white and ryan smith
and this video in 18.0 is going to be
about
setting the context for 18.1 and 0.2

whether participating in these
discussions with the authors
if you're listening to this before those
events happen or contextualizing
um in sort of a bi-directional way for
listening to those later
discussions so if you're in the active
inference community this might be
exposing you to some new questions and
perspectives
if you're from outside the active
inference community or you're not
familiar with it
then this will be highlighting how
active inference is an integrative
framework across a few different other
ideas these dot zero videos are just
introductions to the ideas because you
could take a whole course of learning
on any one of them so they're just
little suggestions and hints related to
the ideas
and where this paper is going as far as
the punch line is that we can build on
this
predictive global neuronal workspace
model
and think about it in being implemented
by active inference
type models and get new insights
predictions and hopefully explanations
and applications
and in 18 we're gonna pretty much go
through a lot of the paper
as far as their goals and the questions
and many of their
copious figures and it'll be good to
talk to ryan and christopher when they

join for the stream
for 18.1 and 0.2 we're going to be
discussing this paper so save and
submit your questions and get in touch
with us if you want to participate
okay so here is the paper predictive
global neuronal workspace
a formal active inference model of
visual consciousness before we
read the goals and claims uh stephen
maybe you could just say hello and
introduce yourself and
what got you excited in this paper just
personally before we dive into like the
paper itself
thanks daniel yeah well this this
my background is that i'm doing a
practice based
phd um at the solomon's
institute of applied psychology so i'm
interested in how
active inference can help understand
some of the processes that i've been
working on
for understanding pre-reflective
phenomenological awareness and people's
sense-making so that kind of ties in
to how i'm also interested in um
ryan and christopher's work because
they're bringing this into the
psychological domain
but also bringing that into the
phenomenological domain
in ways which are practically
predictive and so that's really
really exciting and i'm i see a lot of
opportunities for this modeling work and
i've actually spoken to ryan about this

to try and link in with other ways
of uh people doing sense making etc so
and then this particular paper um
i think is really interesting because
we've got a lot of
exciting um philosophical ideas around
global workspace concepts
in neuroscience and an area that i
i've only partly explored but to have
this integrated into a predictive model
that can be tested at certain scales in
certain ways
and to have people who've got the sort
of rigor and the kind of
the ability to focus down and choose
um experimental and practical
ways to to explore this which vine and
christopher have is really helpful so
um so i'm looking at this from a few
different ways and hopefully
that will be something that other people
will find useful
nice so this is kind of a big
paper and topic so we're gonna frame it
first with just how the authors
represented their goals and claims and
then we're gonna kind of
pull back to just the bigger questions
about the paper
and then we're gonna go through
basically the keywords
so if when reading what the authors are
setting out to do
it's like why would you even want to
travel from a to z we're going to go
to define them so um stephen do you want
to read the first
one of

on six
oh wait you muted
so so goals and claims of the paper here
we've got
formalizing ideas uh first introduced by
how in white and fristen so um
they argue that conscious access
um or ignition which is this kind of
sudden awareness that things come into
our awareness
is is a fundamentally inferential
process
that depends upon a level of processing
of sufficient temporal depth to
contextualize and coordinate
lower levels of processing so that's
actually quite
um has a few implications there
so and so i carry on is that okay so um
then also to build on the previous
conceptual contributions in this area
mentioned above
um they they will be substantiating
arguments
with a series of detailed computational
simulations
these stimulations are based upon the
first principles account
of perception and action selection
offered by
active inference so that that's
again that's one of the real
foundational benefits that this
is testable um and then also
um they end by examining the
relationships between
um the predictive global neuronal
workspace which they've introduced as a

concept
and alternative models and briefly
address potential concerns about
how phenomenal consciousness could
plausibly
be situated within the model that
they're going to present
and however this paper is chiefly
concerned with what
block terms access consciousness
which is defined as the availability of
information for verbal report
voluntary action and executive
processing for brevity
they will use conscious access they use
conscious action
access and consciousness interchangeably
throughout the paper so just to really
you know
that this gives them a practical
starting point to work on
yeah nice yep there's a lot
in these goals but one of the key pieces
is that they're talking about
informational availability in terms of
report
paradigms and then they're going to be
connecting
big questions in multiscale systems and
cognition
to active inference so let's unpack that
a little bit in terms of just what are
the bigger
questions not in the author's words that
we identified this paper
as relevant for so people who are maybe
working in these areas
and the first one uh or which one of

these do you want to just go for first
i might just mention the first one and
then if you could pick it up after that
one
um so yeah basically there just
there is a question really um that
they're exploring
can simulations using generative models
of belief updating
um give you principled schemes
in active influence to um to extend
to other boosts for a predictive global
neuronal workspace so you know does this
lead to sort of a plausible
doing this sort of simulation to a
predictive
approach to this whole paradigm nice
and it's related to the second question
which is what are the unique predictions
and explanations of active inference
so it's the first question is kind of
asking how can we use the simulation
the generative model how can that be
used to get
to unique evidence or unique claims or
unique experimental design
and then something that's related to
active inferences role
is how could this reposition the
distinctions that people make between
access consciousness and
phenomenological consciousness
so access consciousness as per the
previous slide
is the informational accessibility
phenomenological consciousness is the
awareness or the qualia
and an example of something that's in

access consciousness
but not phenomenological would be like a
subliminal
sound because you're it's not
experiencing it it's subliminal
but it can influence somebody so that is
within their access consciousness
because
informationally it leads to differential
behavior
or the planting of an idea but it's not
phenomenologically experienced
so it's not just one basket with what
um is being met with consciousness yeah
stephen
yeah and so to clarify a little bit on
that as well is
that they they they are using this idea
for this
access consciousness whereby um you can
access it in a way that you could report
on it
um like you're saying and it may be
there
like i said as a qualia and it may
be transiently available but you may not
be able to access it from a reporting
paradigm so they
the way they frame access consciousness
is to really give a temporal
depth and a sense of how stuff comes
in so the idea of access consciousness
is either accessible or it's not
whereas phenomenological consciousness
is kind of more of that
flux which is when we think of
consciousness it's hard
there's not really something you can put

in a box whereas access consciousness
you can't put it in a box either but
they're more able to so they can model
it
right it's beautiful yes they also
are setting out to address how the
active inference
model can integrate different ideas like
bayesian brain
and global neuronal workspace in
neuroscience
and combine that with more psychological
applications
and also they're asking in this paper
how can they
um perhaps integrate with other existing
active inference models
to sort of extend other work like
related to the global neuronal workspace
and also
bringing it into a specific visual
modality and experimental task
so we had a bunch of questions making
dot zero
as well so what is just one or a few of
the questions that you just want to
raise and what
why'd you erase it
some of the questions that that came up
for me was um
you know just how does this all um
fit together into what's potentially in
the future so i'd be really interested
to get ryan and christopher's sort of
take on some of these
distinctions you know um just like you
were talking about that potential for it
to go into

multi-sensory integration and beyond
so how does visual consciousness or
awareness of that
differ from general awareness
and how does the ability of the
predictive global
um neuronal workspace um give these
detailed simulations and
resulting predictions so what's the
potential for that
in other fields and sectors okay so one
of them might be to help influence
psychological perspectives on people's
professional practice
um as with people are starting to make
um even what it means to take a
perspective
how much is that access consciousness
effectively
when i start to bring something to my
perspective am i bringing something to
my access consciousness
when medical understanding of conscious
awareness
or when people start to understand about
self-reporting in general and how people
know and people who are working with
knowledge and bring things up
because a lot of knowledge workers are
using interviews and reporting paradigms
um and there's a big question about well
how do you deal with first person
normally it's first person perspectives
which is one thing
and then there's first person experience
and how do you
even get those two resolved
in in some sort of coherent way so i

think that that
that that's something that would be
really interesting to to
hear more on nice good
thoughts and we're just always curious
about where active inference
comes from and where we can apply it and
learn it
why does it really matter well there's
sort of a theory
arm and an applications arm the theory
arm
is that this paper is exploring some
experimentally tractable unique
predictions
of an active inference type model
and showing how it goes beyond bayesian
brain global neuronal workspace
predictive processing it integrates
across and it goes beyond
several of these theoretical
neuroscience modalities
and then the paper is part of a broader
network of research and work by
ryan and others that we'll hopefully
hear more about just
as to how could this shape the kinds of
clinical
and community psychology practices that
people are
performing because this is related to
all these types of uh
topics so any thoughts on this before we
go to the abstract roadmap
yeah i suppose it just comes back to
this idea of ways of knowing what it
means
to know and be aware and have awareness

um that that actually is often sort of
implicitly sort of in the background but
this gives a way to make some things
more explicit which i think
could be really valuable
cool so the um
abstract just because i think we'll have
enough to actually talk about with a
paper
people can pause it to read the full
thing they're
in the first part of the abstract
framing the need to move beyond
the global neuronal workspace model
in the second half of the abstract they
report what they did
which is demonstrating their integrative
model which we're going to be able to
look at and explain what they actually
did
and they find that that model is able to
reproduce
a lot of other findings from previously
non-unified experiments and then
they close with a discussion on how they
can
build upon previous work in this
integrative and
active inference style and where that
leads to as far as predictions
so that's sort of reflected in their
roadmap in that they give a basic
broad introduction then specific
theoretical
backgrounds on active inference and deep
temporal modeling
they then spend the bulk of the work
presenting and many figures presenting

the results
and recapitulating several specific
phenomena
that are important to reproduce and also
impressive to reproduce
in a simple framework and one where we
can see all the parts
and then they close with basically a
discussion about
where their work is fitting in relative
to other work
and concluding summarizing and stating a
few
next steps so how did that strike you or
what would you add to that stephen if
anything
no i think that's a good summary i think
in some ways we might be
the authors would speak a bit more to
how that taxonomy came about and
there's some nuances in there in terms
of how and why they selected what they
selected and
um which um would be really interesting
to
to have them really step through um but
uh
and um but overall the
the way they've modeled this out into a
very practical
um real world kind of context i think
shows that potential and so and then
um what does that mean for this temple
structure
then links back to other work around
temporal
depth which is you know the kind of
since i think it's

2016 has been what a lot of the active
inference
work has certainly in the modeling has
really been
um yielding is this idea of temporal
depth and what it can mean
so that'd be great nice great point
earlier models of active inference were
sort of like
instantaneous or one-step
models and a lot of the ongoing research
is about the integration
of multiple scales or multiple course
gradings of temporal dynamics
so here are the keywords the paper so if
someone was listening to it from
any background and was curious about why
does it matter to connect some of these
things or why
why is it important or how does that
implication arise
from integrating what i've previously
people done if not with this integration
that's where the keywords kind of come
into play and we're going to go through
basically a slide for each they are
computational neuroscience
predictive processing bayesian brain
hypothesis
active inference consciousness visual
consciousness
global neuronal workspace and the
predictive global neuronal workspace
so let's start with computational
neuroscience
a lot you could say about this a lot of
people's department
and research but we can just look at it

pretty naively as the intersection
of computational approaches and
neuroscience research
so computational could mean using
computers
which a lot of things are in nowadays
using computers could be descriptively
modeling statistics
or it could be generatively modeling a
system like with a simulation
but also the computational approaches
might include using computational
metaphors to describe systems that might
be considered computational as well
like computationalist on the other hand
neuroscience is about studying
brains but increasingly non-neural cell
types in the brain and the body and the
niche and society
so all these different aspects of
behavior in the brain
and it uses all different kinds of
approaches some of which
are computational which brings us to the
overlap which
can look like a few different things it
could look like computational brain
metaphors the brain is a computer
or it does information transfer of this
type
um it could look like doing descriptive
or generative modeling
on computational representations of
neurological data
it could look like brain computer
interfaces so that's sort of
computational neuroscience
and stephen i know that you like kind of

saw where that could go
beyond where most people would think
about computational neuro
do you want to mention it here on the
next slide
um maybe on the next slide and just
i suppose in some ways it's just to try
and highlight some of the
thought practices i like to go through
to to get my head around what
um this means and how i can bring it
together with
the active inference process so the the
the challenge you slightly have with
active inferences and predictive
processing that goes with this um is
that
um it challenges the computer metaphor
that we've kind of got embedded
of how the brain works either the model
or the thought that we sort of bring in
signals and then we have a computational
awareness we think it and then we do it
it's this sort of
open system so these predictive
processes and this cybernetics
so it kind of when we talk about
computational neuroscience
in relation to the systems neuroscience
and non-linear dynamical systems that
active influence is part of um
it's it's a bit more of an open um
kind of process approach
so i suppose that to for someone who's
trying to understand this
i think i found it helpful to start to
to think about the the computation
computations that dynamic process that

develops over time
when it's used in this sort of context
so it's it and those sorts of ways that
these models are
which is different to kind of
system dynamic models which are much
more like
flows on something which is defined it's
this kind of
evolving process and of course the free
energy principle then ties into that
because it's all about being able to
work with those dynamics
so um i don't know does that make sense
what's there
people can seem to let us know if it
makes censor helps them
i think that the non-linear dynamical
systems is a big topic
one way that that has been
operationalized
in neuroscience is the bayesian brain
hypothesis because bayesian statistics
is often used to model non-linear
dynamical systems
so if we're thinking about the brain as
a non-linear dynamical system
it makes sense to think about the brain
as a bayesian device
and this is from the 2018 interview with
fristen and martin and myself and
in this paper we summarized
and characterized the bayesian brain
hypothesis predictive coding and the
free energy principle
and asked how are these uh different
models
similar or different from one another in

fact he had suggested that these three
frameworks are variations of the same
basic mechanisms
so if you're wanting to learn more about
the background of the bayesian brain
and how it differs in specific ways from
predictive processing free energy
principle active inference etc
then read his response here
or read the interview because um it can
help with
some people's understanding let's just
go through it at an overview level
predictive processing is specifically
looking at how
error is propagated through systems
which
can be hierarchical or must be
hierarchical and therefore
are amenable to a bayesian hierarchical
modeling
so in this sort of diagram on the right
it shows how
at the very bottom level you have
muscular output action and sensory input
and then what's happening in this
cascade
of hierarchically nested systems is that
predictions are coming down
and prediction errors are moving up but
it's not a spatial thing it could be in
or out or it could it's just
about how it's laid out and predictive
processing
turns out to be a way to get
somewhat effective bayesian-like
behavior
because it's even how we fit some of our

algorithms we just start with the prior
and then we inch it closer
depending on the error signal so
learning based upon the error signal
is a really common algorithmic maneuver
and so predictive processing
in the bayesian brain are sort of like
compatible theories because the bayesian
brain is just saying
it's going to be something bayesian
predictive processing
is a theory because it makes predictions
that are falsifiable about
what is being represented in the
communications between levels
and namely that you have prediction
going one way and prediction error
heading back the other
way anything you say about that before
we continue
no that makes sense so the only thing
i'd add is that what's going to be
mentioned is how there's um this
temporal
um depth piece starts to make so you
could have like a level
of processing goes to another level and
that
that's something that but the same basic
principle is driving it
so yeah cool so here is a really
informative piece
from the first in 2018 interview
fristen wrote the bayesian brain
hypothesis per se
does not trouble itself to commit to a
particular process theory
other than requiring implicit beliefs to

conform to bayes rule so similar to how
free energy principle
doesn't have a specific process theory
that's downstream it's a more like a
general overarching does it conform or
not to this
principle the bayesian brain hypothesis
is a corollary of the free energy
principle
and is realized through processes like
predictive coding or abductive inference
under prior beliefs
however bayesian brain is not the free
energy principle because both bayesian
brain hypothesis and predictive coding
are incomplete theories of how we
inspire
states of affairs and then here is so
interesting
where he wrote it's this inactive
embodied
extended embedded and encultured aspect
that is lacking from the bayesian brain
and predictive coding theories precisely
because they do not consider entropy
reduction
through action as well as through
inference so it's all
well and good to have the inference
engine that's driven in this
thermodynamic way
towards entropy reduction but it takes
on a whole other level and integrates
with control theory
and action in the niche when you go from
saying
well yes the brain is a bayesian
statistical inference device

for action and that's really one of the
key pieces of active inference
which is why we went to it here so
before we
um this is the definition of the authors
in this paper that's what they wrote
so stephen where would you say active
inference fits in or what do you have to
say about that well i like this this is
a really helpful slide well there's a
couple of things that come up one thing
is
the fact that it talks about um
abductive inference
and it's talking about this idea you
know predictive coding and um
you know ryan smith and christopher
they're talking about prospective
sometimes ideas of you know we're
predicting how
things are going to happen well that's
really interesting because most
research in terms of like social
sciences mostly
you know retrospective or inductive
or deductive and abductive is
this abductive emergent thing is is
actually very
um hard to get at but actually is is
showing that this predictive
rather than retroactive knowing
is what's driving everything um so i
think that
that in itself is quite useful to think
about um
yep nice it's interesting that he
mentioned it there
and that active inference how how does

it implement
potentially um some of these cool
aspects
that are lacking from the bayesian brain
so where does active inference build on
bayesian brain approaches here's what
active inference is
so the way that the authors uh ryan
smith and christopher white
wrote it in this paper on the first
sentence of their primer
is active inference a quarterly of the
corollary of the free energy principle
is the first principles approach to
modeling approximately bayes optimal
behavior
so you can look at their primer in this
paper
which is really awesome as well as
they're very extensive tutorial on
active inference and model stream
number one point one through four so we
just
really appreciate the clarity that ryan
and christopher bring to this
area but basically what's happening is
active inference
is allowing us to formalize the
process of the environment giving data
through sense
to the organism which in the generative
modeling of that data
which is like bayesian modeling but
that's inference
and then it connects it to
external states again closing the loop
via action as well as internal dynamics
or endogenous fluctuations on either

side of the interface
only by defining the interface do we get
this kind of a clean separation
so that's where active inference allows
us to say yes it's bayesian in terms of
how it recognizes data
and it acts in an ecosystem that also
has other agents
that's where active inference comes into
play all right
so now to sort of switch gears to
consciousness
and visual consciousness so let's just
start with sort of the obvious
what is consciousness question and just
we're just talking about this paper so
we're not talking about like
something that's you know some sort of
theological
question remember that it's access
consciousness it's
self-report paradigms and you can read
more about their perspective on that in
the limitations and
multiple other places where they really
are clear about the pros and cons of
using
visual modality so it's not olfactory
consciousness it's not a sound it's not
multimodal
they're doing a sense specific task
and it's reported awareness so
what do you think about that stephen or
in your you know thinking or learning
about
consciousness where would this sort of a
narrow approach
figure it

yeah well i think that um you know when
they talk about the pros and cons
um you know the the the
pros are that we are actually able to
generate data through reports
and it gives us a way into
this whole area and by
the sort of the detail of
how the design has been done it it
negates
and makes that negate some of the
challenges and makes us get
some workable insights um and
one of the cons is that or one of the
downsides that they're facing is you
can't genuinely
report being conscious of nothing you
can't say you
i'm no longer conscious of any of that
thing
so to speak and they acknowledge that
there's always something in your field
of
vision when you're being asked to
self-report around if something's
present or not present
so um but so they've had to do some
workarounds to get the results that are
usable
but um that that would be something they
can speak to better than me
cool i agree it'll be like um what are
the authors big
motivations what is their long-term
vision for consciousness research why
have they chosen vision first
what would be the next most tractable
sense so definitely there's a lot of

questions to have for them and also this
is a great note from them
brief note on phenomenology and the
phenomenal
consciousness access consciousness
distinction so again i just
we kind of hit this point because it's
one of the it's one of the key words of
the paper
is consciousness and a lot of times when
uh
we see scientific research on
consciousness so called it's just
it's always critical to be clear about
what's being discussed
and the authors thankfully here really
are clear which actually leads some
people to be almost disappointed it's
like
oh but what about the big phenomenology
it's like
maybe we'll get there or maybe we won't
or you could have any story you want on
that
but it's almost like if the sales pitch
is too much
hype then people it's what they expect
so then when there's a more narrow
focused paper they act like it
introduces
all these critical challenges that are
just being
glided over by a paper that doesn't
define consciousness or doesn't
go so specifically and carefully into
the distinctions between access and
phenomenal consciousness because not
everyone will

and uh yeah yeah i like that
the the way they talk about this um
they acknowledge this question about
this threshold and start to
sort of decide on where they're going to
fit in with that i think is really
valuable because
um this idea of
or the question that's sitting in the
background with phenomenological
uh consciousness there when they talk
about this moment-to-moment
sensory flux and in a way it
ties to the um chris fields
work that we were doing with quantum
contextuality potentially
um is you know what's this kind of flux
idea that's in the background and when
does it get
tripped over to the contextuality of
awareness so
um and that's that but just to show it
linking to some of the other papers that
have come up
and then for this you know it shows you
the importance
of the experimental design and
the modeling simulation design and how
the way the design
ends up evolving itself
kind of gives us some insights of how
you might structure
our understanding of consciousness the
sort of questions they had to ask
and answer to build the model in itself
is quite interesting
nice yep so they're
um really clear about that and we can

talk more about it
let's go to the neuro side so we kind of
talked a little bit about the
philosophical question of consciousness
and the systems dynamics and control
theory question of active inference
but then there's this neuroscience side
and
area that they're in and that's what's
also moving towards applications in the
clinic for example
and what we can do here is present some
of the early
and latter citations related to
the global neuronal workspace global
workspace theory gwt or
there's a bunch of related ones sorry if
the acronyms are not always consistent
but
um what the global neuronal workspace i
guess is
probably what we should use and
predictive gnw
are there integrative models that are on
one
hand for neuronal function so
through philosophically neutral
frameworks like cascade theory or
network theory you can describe
how you get patterns like ignition or
cascades through networks or propagation
then it can be applied and say ah what
if
that ignition is so-called consciousness
access consciousness there's a strong
argument you can say look if the data
doesn't get to the other end of the
pipeline it's not going to get there

so if it does get there it had to have
propagated through a critical system
so that is the easier bar
and then the harder bars is that
information propagation
equivalent to phenomenal consciousness
experience qualia
and the answer as we know from
subliminal stimuli is just no
it's not but that's what people often
want to get at
addressing which is how does the brain
quote make consciousness
or what brain states are associated with
quote consciousness meaning awareness
in our first person experience which
even under the most generous vision is
just a part of our
total system what do you think about
that stephen
yeah i think that's a good and you make
a good point there and
this loom from the sort of more broader
workspace to this neuronal workspace
then brings in
um the actual understanding which again
ryan and christopher have of of
neuroscience and the way that
the the firing happens so that gives an
insight
and also the thing that i think is
interesting is
that they're talking about consciousness
as being this
belief um updating or prediction that's
fairly uh on off like it's like
things come in they come out of
consciousness um

whereas in other cases that might be
more of a gradual
flux of awareness there's something
going on here which because that flips
in and out of consciousness that gives
you a way to use
um belief updating schemes
to model it and i'd be interested to
hear how
they see that yeah nice good question so
here's kind of a
classic neuroscience diagram from this
2005 early paper
left side is the single neuron scale you
have receptors in a membrane
and different currents that are
happening at a very very
micro level meso level with circuits and
then more of a macro level
with the activation that's propagating
across regions of the brain and then
another
way that was helpful was uh looking at
this
i think it could have even been 1998
paper
the global neuronal workspace model of
conscious access from neuronal
architecture to clinical applications
so lest we think that it's a new idea to
apply neuroscience
in clinical settings or think it's too
soon or too
late it's just always going to be sort
of what people are recalibrating too
and in this figure it's connecting
sort of the uh multifaceted
attributes of the self and consciousness

so again used maybe vaguely maybe
detailedly
like long-term memory of the past
evaluating systems of value attentional
focus
regimes of attention as we would say
perceptual systems
in the moment and then motor systems in
action in the future
so it's like if all those things are
being connected and
what connects the sound in the ear
to the memory to the action pattern like
if someone says
march what connects that and the answer
is
something like a system at criticality
that propagates from a sensory modality
and if we're a quieter it wouldn't have
been heard it wouldn't have propagated
if it's the wrong language
it doesn't propagate so it kind of only
catches
when it's all aligned with the rhythmic
oscillations and the prior state of the
network
which is equivalent to constituting a
bayesian belief network
so that's sort of the global neuronal
workspace model is it's going to be
integrating all these features that are
kind of disparate
cybernetic features memory past present
future self other
thinking through other minds all these
features of neuroscience that everyone's
trying to
understand and model and put that in a

network science
frame so that maybe it could be
brought into the clinical applications
just like their title is getting at even
20 years ago
or other avenues like how do we discuss
neuroscience with the public
what kind of metaphors will they receive
is it like a network that
activates or an avalanche that falls or
a program that halts
different ways that people think about
systems neuroscience
any thoughts on that before we can
continue yeah i think just generally
speaking this
this idea of this integration um is
really useful when people are thinking
more broadly about um you know often
people are thinking about us as a system
but you people fall into this trap
of i'm a kinesthetic thinker or a visual
thinker
you know some of these kind of folk
psychology systems things and
nlp and that and this approach gives a
plausible way to understand what it's
it's it's all one integration and
it's not that you have someone going off
and their
visual system is in one part of the
brain and this piece
you know and in a way this is permeating
how culture kind of thinks the brain
works
so and then bringing that in with the
free energy as a way that it could
actually integrate and it could be part

of that
and now this prediction um dynamic
it is quite um
it is quite possible that this could
start to be
more understood more generally as a
model that
is how we sort of work you know as to
the
uh different styles of learning or
sensory modalities it could be that
some cognitive diversity is such that
the same
symbols on the page will ignite
something
that the audio doesn't depending on
where they can
have their regime of attention which is
a culturally
inherited and co-created artifact so it
allows instead of just saying well it's
left brain right brain or it's this kind
of person that kind of person
it's about the dynamics of the system
and oh would it be possible to do some
sort of exercise to sensitize you or to
allow you to perceive
some dimension that your brain hadn't
seen just like there's a good
music or a good wine or something but
it's something that
is encultured and trained so let's get a
little bit specific into the
neuroscience
before we go to the figures and that's
just to talk about this erp
or event related potential and the erp
is a pattern that's kind of like a

collective behavior
of an electromagnetic field it's based
upon eeg
measurements which are
electroencephalograms and those are just
like electrical
measurements around the scalp usually
but it can be other places too
and each of the little leads each of the
measuring
points on the eeg is a reference channel
and then what happens is you could be
taking these time
series of the electrical potential at
these different reference points
and then using techniques like spm and
other approaches
to reconstruct you know generative
capacity
the electromagnetic field that gave rise
to that
data distribution of event leads
references channels then
you can align time series to events
like if you clap your hands 10 times
separated by 30 seconds
you could take that big time series and
then align all the claps
and then ask is there like a coordinated
probabilistic not every channel does it
but when you align it you can see that
yeah there is a dip and so that's the
erp
is when there's this spatial temporal
pattern that might play out over
tens or hundreds of milliseconds as
shown here
and in different brain regions it's

known to have a canonical form
and even diagnostic value in certain
situations
and there's a lot more to learn about
here fristen's work
buzaki many others but it's a big area
but that's kind of what it is it's like
a pattern
of space and time in an electromagnetic
field in response to an event
any thoughts on that because it kind of
comes into play for the paper yeah
nice and and i noticed they i am this
event related potential so
yeah again it's giving you a way to to
think about what happens when so you've
got
p one p two and
p3 they talk about p3 more i noticed in
the paper
um there's this kind of main sort of lag
but it gives it gives a sense of this
temporal dynamics that are played
there's an idea that something happening
which is more cumulative maybe takes
more
time to go to
register and that
but it this this this model gives a
plausible way where
that could hint that some of the faster
processes are actually
sort of more closer to this flux and
some of the others
are closer to this awareness
your consciousness yeah it's a lot like
the pq
rst system on the heart

ekg and so you can say oh there's long
qt
interval it's because there's sort of a
normal shape
with variation and then if something is
outside of a certain bound you can say
this sort of interval is too long
and then you can have a model of heart
cells and if you had a model of heart
cells with just the channels
and you're getting the range of
expressivity absorbed observed in the
biological system it's pretty meaningful
so let's look at figure one the first
half of figure one i guess figure one a
we could call it but we'll have figure
one b2
this is kind of the classic active
inference
graph this is one of our favorite
figures
and it's good that they put it in all
kinds of papers
what it describes is the relationship
between at the starting at the bottom
observed
measurements in the world that are
mapped through uh
matrix a to hidden states of the world
like is it actually light in the room
is the hidden state and then you get
photons on your eye and then there's a
pretty clear mapping there but a
could be probabilistic as well so that's
about states that are hidden
to observations through time um that are
observed then this
is actually a second level model where

each hidden state it's evaluating
what is going to be the state outcome
mapping under
a future time series so this is a little
bit not the basic basic version but
hopefully you'll have listened to a few
other um like their model stream where
they walk through from really
the fundamental which is just the state
outcome
mapping so go to the model stream if you
haven't seen this before at all
then um d is the prior
that plays into these second level state
estimates and just like the b matrix at
the
lower level defines the transition
between hidden states
the b2 as opposed to b1 second level b
matrix
describes the transition frequency and
patterns of
the states at the higher level so this
is kind of like that nice
nesting you could have the higher level
state changing and then lower level
states changing
what does influence the higher level
state transition matrix pi policy
so that's the innovation or one of them
of active inference
which is that instead of just estimating
how states change
hidden states change through time we
estimate a conditional
related to how our policies influence
states changing through time that's the
pie coming into b2

our policies are selected
from our affordances and they're sort of
differentially weighted
in terms of their instantaneous uh
attractability
in terms of this g functional which is a
expected free free energy minimization
of policy
down here g of pi not just some abstract
g floating out there in space but
of a policy and it takes in
arguments like c the preferences
and c is the preference vector okay so i
think that's all the letters on here
good any thoughts on this steven or we
can go to this figure one b
no i think well i think this is a one
thing i've noticed is also they've got
at the top the variational free energy
which is kind of the
the broad kind of dynamic that flows as
you mentioned and like you just
mentioned
it's the when you start to got the
policies involved
um you've got this expected free energy
which um
is more tied to
how that might relate to particular
actions for the organism
um so i think that makes and also i
suppose the other point that's there
you mentioned there you know the b1
matrices
it it could almost be assumed that
that's a that's like the flux it's it's
kind of
it keeps churning over and it will it

will
it will work out it will come and start
to populate its values
whereas these higher levels you have
your priors it's
in you know it might not be that
important where my retina is
in its state at this moment
it will just populate and start to come
to some equilibrium whereas what i think
i'm seeing
my prior might be at these higher levels
a more
significant parameter and that's
reflected in
that model cool very nicely said about
how
our priors or system priors even to
remove the whole
awareness consciousness thing system
priors are semantic
it's like i expect i'm a veterinary
scanner or i'm a check scanner
at the atm i expect a check
it's like it doesn't expect the pixel to
be in this spot and the green and the
this way to be here
it expects something semantic and part
of the challenge
with talking about observations and
making meaning from data and sense
making from large data
and scaling sense making is that the
bottom level data is almost by
definition
meaningless especially with respect to
the larger level semantic
question so yeah there's parts of the

check that are sub meaningful
but the pixel can't be too meaningful
otherwise you need to do a whole nother
model to really break that down
so it is interesting that the d prior
enters at the second level state
but there's no d1 so maybe it's not
shown or maybe we can ask where that
is coming from let's look at yeah maybe
it's not
it's not necessarily so important for
the modeling and
so that that in a way this this is quite
useful for
showing i mean in some ways they say
that's where autism can have challenges
because
that there's not that ability to
know where to put your priors and and to
to resolve things so
um just one other point in here is they
don't
in the expected free energy for vision
because they're looking at sort of
salience of vision
or fragment the idea of risk
which is sometimes shown in that
expected free energy is not in there
because i suppose for this model that's
not really
it's not relevant for the vision system
in the type of
way that they're exploring consciousness
so it just shows again that the way that
the equations or the terms can be
constructed is related to the nature of
the way that the what's most relevant to
the type

of dynamic that's being looked at
and i think that's quite interesting
right it's not the free energy
of like the system with a no function in
mind
it's a calculated free energy that is
related to
policy as implemented by some actor in
the space
so it's not just like we're making one
general free energy principle model of
some
system we're doing task specific
specific behavior
and i'm sure that ryan and christopher
will have a lot to add on that
so here's the second half of figure 1b
or figurehead one
and it takes some of those letters that
we saw
in the first part of figure 1 a b c d
and then also adding in e i think down
there which is the kind of affordances
prior which is not here but it connects
to pi as well
so there's a few other aspects and then
it maps it onto
parts of the brain human style brain
that are tractable or plausible
or where the architecture of how those
brain regions are connected or
functionally linked through dynamic
causal modeling spm
so if there's a functional or an
anatomical connection between these
brain regions
with that directionality it seemed to be
some might think of it as confirmatory

or supportive evidence
for that biological systems do free
energy principle or active inference
another way of thinking about it is it's
the computational
realization of a process that
surprisingly or unsurprisingly can be
modeled with active inference type
models there's a few different ways you
can think about it but they're kind of
laying out this probabilistic
graphical bayesian network and they're
kind of laying that out onto the brain's
neural architecture and there's a lot
more papers that go into detail here
any thoughts or ready to continue
yeah i think an interesting question is
like where does the body come into play
like this is one level of the nesting
but
how far is this model towards
just basic group communication models or
some other type of modeling anyways
let's return to the sort of
specifics of their experiment they
included figure one i think
to be just very clear about active
inference which is great because it
helps the paper
be accessible to those who are outside
the field and also
everybody in active inference can always
go back to the basics and the terms
because it just always helps us here's
what they do
in the results the first part and then
we're gonna go look at figure 2 and the
task

so results section 3. as a proof of
principle we first simulated 200 trials
so they're going to be having a
behavioral paradigm where they can have
a fully
in silico fully simulated experiment
so it's going to be like an experiment
that a person could do but they're going
to totally simulate it
and some of them are going to have
square present
and some obviously won't and then
there's going to be model parameters
that relate to attention and model
parameters
that relate to stimulus strength so off
the bat
you might imagine okay well if the
stimulus is stronger like the sound is
louder
to some point it's going to be easier to
say yep i heard that sound
so if it's a bold line versus if it's a
tiny faint
you know gray on gray situation
and then there's also the attention so
if it said oh pay no attention to the
middle look away and then something
flash in the middle you're less likely
to see it
than if you were to be looking at the
middle so that's the kind of an
experimental setup that they're going to
be simulating
completely in silica here is what
they're going to show in figure 2.
there's this sort of visual pattern at
time point one

and this is what also allows them to use
discrete
time models like as currently shown
the active inference model where it's
like t1
is here t2 is here and t3 is here
so they can actually get a lot of power
out of this sort of like checkpoint
type almost kairos driven analysis
really and then they show an ambiguous
stimulus where it replaces these
diagonal
lines with partially with a square
and then it flips back to the bars the
agent is then required to either
uh say yes i saw a square or
no i didn't see a square so that's the
sense
and here's the action so if anyone has
made it this far
and was still like wait but they didn't
explain you know consciousness
of this system or of theology it's it's
not
this paper this is only about this task
paradigm
any thoughts on that or continue
yep all right so
they um this was just useful in terms of
from their thinking in their discussion
for those who want to read a little more
about why the time scales are the way
that they are
okay figure three is a bayesian network
description
of the generative model with arrows
showing dependencies between hidden
state factors and outcome

modalities okay so this is kind of like
at first i was like wait what is this it
looks like a
an app or a game or something like a
network model
but it is actually so at um
the level one
let's look at these are the hidden state
factors these are the report
would be i'm waiting i haven't reported
yet which is actually a state
seen or unseen so this is sort of
one of the hidden state s factors that
we're going to fit with data
then there is um the trial
phase which is the time steps of the
model
and then there's going to be the uh sort
of visually
differentiating features of the model
so it's sort of like you know round one
inning one
first out it's kind of like identifying
where you are in the model as a state
in the model which is something that was
really important
in this framing of the model so check
out the model stream because it matters
if you're in
trial two row three or trial four row
two or something like that
and then at the very bottom level is the
square or the lines
here there's also this red or black
thing so i'm not going into that but
it's whether there's the square or the
lines being seen
and then words i uh silent or i

see a square square didn't see anything
okay
so it's sort of like the generative
models if they saw it or not
so we need to have some integrated way
to combine the trial phases
and what is actually being displayed
which we control in the experimental
context
with their hidden
perception as to for example the
sequence of two different things
so if it was first the square or the
diagonal and then red or black
you could have four different
combinations and if it is
held across trials you had to have like
been aware of it
so the thinking would go like i'm going
to read you a color and then a name and
then you're going to tell me both in a
row
and so that's a report that they're
going to be able to say
and then that's the attentional mask
because that allows them to say
not report on the color but report on
just the square being there or not
so it's kind of like everything in this
experiment
is like locked down not that everything
the person's shirt color
and how they felt about it not that that
is all explained in this model
but it's sort of modeling the input and
the output and then leaving the rest
actually for context
so we can hear more about that from the

authors anything on figure three
yeah and i think they can speak to that
more i think one thing is what's
interesting is they they use this
these different words rather than just
like
red or no you know just basically saying
red when you see red or black when you
see black
they had a rationale there to separate
it out
so that it helped them have a bigger
time window
um but i'd be i think they could speak
to that better than i can so
yep nice the level two outcomes are
level one hidden state factors
so the higher level is getting um
connected to lower level in this
way and then also if you really follow
some of these lines like weight
only maps to silent and that only maps
to the observable silent
scene maps to different things in unseen
they both are i
see and you know could be uh
that's it like i see that's where the
divergence point is they're both about
a perspective and perception and then
a square that's only for this thin dash
for scene
or i didn't see anything from the unseen
condition
so it's kind of showing how even the
lexical tokens
are mapped from probabilistic estimates
of higher order models and that also
relates to this quantization

and this quantum nature of communication
and it's like
you're forcing them to say i saw it or
not it's not a slider how sure where you
saw it you could design that experiment
make that model but in this paper
there's a lexical interpretation of
their actions
because they're only given the option
yes i did see it
no i did not see it so by constraining
the experiment
you can uh do things again
leave it up to someone to debate whether
that is uh cutting too much away
all right so four is split in half
and here's where they sort of are just
proof testing
their simulation that's the foundational
simulation
and this is just to check
some of the basis of their model and
what we can see
on this left one is that when the square
is shown
so here's the four possible hidden
states it's like four horse racing lanes
and we can see that there's this gray
that's
the red one is never even on the table
because they haven't even brought that
level into play yet
but there's this gray competition 50 50
i'm not sure
that's just reflecting the prior of
whether it's going to be the dashes or
the squares because remember
did you see it or not and not is going

to be the dash
and then the stimulus occurs and then
there's rapidly a convergence towards
yep i saw it
and that convergence is associated with
a transient period of elevated firing
rates of
the first level of the model so that's
where we're going to get some of the
cross
experimental modality work because this
isn't just a bayesian phenomenological
analysis of probability it's actually
linked to
a mechanistic process model so if you
found a brain region that were firing
that way
and removing it did the same thing that
removing this did
all of a sudden you'd start to be seeing
a lot of model reification
and then they also tie in the
probabilistic component firing rates
the first level firing rates in the
self-report and then conversely
on the square absent condition it's the
same thing it's gray
and then there's no transient period of
firing
and then it just converges towards not
seeing anything so it was like it was
waiting it was waiting
and then it converged upon not seeing
anything so that's figure four that was
just sort of like because they're like
oh yeah after
three four five like i'm waiting i'm
waiting waiting uh no

just didn't happen so that was one
now let's look at another um a little
bit more nuanced way
to look at that in five so here there's
two
axes it's kind of like a punnett square
or eisenhower matrix a two by two
and it it's called a four-way taxonomy
so just like we kind of hinted at
earlier
you can tune model parameters to have
low or high endogenous attention
and then you can tune model parameters
so that the stimulus is like shriek
is weak versus strong so that would be
like thick versus thin line or could be
rapidly flashed versus not
so you could imagine if it's a weak
signal
and the attention is absent very few
are going to see it if it is a
strong signal and attention is being
paid you'd expect a higher amount to see
it
so empirically what you do to make the
attention increased or what you do to
increase the signal strength
that's a detailed question but their
model
is doing quite a few things here and so
it's a higher high energy high density
representation because they're showing
that basically
they do recapitulate that pattern with
high percent seen self report and high
correctness rates
with an archetypal firing rate at the
first and the second level

that are reminiscent of erps
and the erp p3 looking like
arises at the second level so it's not
like
the data have a structure that give rise
to the p3
it's like the p3 is an emergent behavior
of something that as you're saying is
more like
forming and unforming at the lower level
in the semantics and then the response
that we actually observed the system do
is this higher level
and then it's absent in the other three
cases
and in those cases the accuracy here's
almost 100 percent
in this it's 58 47 61. so it's about
50 50. and remember they have to make a
forest choice
and they're not always sure their
confidence isn't 100
so that just shows that like you get
confidence
and experience and firing rate and
second but not first level erp dynamics
in the high attention high signal
strength condition
very clean finding and then you can
disentangle
attention and signal strength and then
just show that like whether you pull
one away or the other you destroy the
erp
you ru you reduce the number of people
who the simulation runs
that said they saw it and it reduces the
accuracy

to a large extent so that's like a
big finding of the model
any thoughts on that yeah thanks that's
really good to go through that i think
um
because there's a lot in this diagram it
can be a little bit uh
to first see and i think just looking
at that you know first level and
noticing how
all four quadrants have got this parry
stimulus time this kind of
first level firing is present like you
just mentioned
in the in the black graph and then you
know only on that
bottom right that you have this very
strong so this this does indicate that
when you have sufficient signal
in relation to attention bang
you know it's it's it has a certain
on-off-ness to it which is
kind of what they're trying to show very
nice to connect it to the ignition
model of the neuronal workspace which is
like
yeah if it's loud but you weren't paying
attention sufficiently
then you're not gonna or the other
quadrant it's like it doesn't ignite but
when you get both together when you get
somebody
looking at the screen and it's the right
stimulus
it's not just a little bit different
than them not looking at the screen
it's so different it's qualitatively
different it's a categorical difference

and so that is something that's
reflected categorically
in this model flat versus not flat now
it's a simple
model but no it makes me
but yeah i don't know it's a simple
model but this is like a lot for a
simple model to show
i was just going to make a little
anecdote there and say it's sunny
slightly to the side but when my
daughter was um i think she was like
nine months old we had the fire alarm go
off
and it was literally like ear breakingly
loud she was sleeping through it all
and so you know that i know it's not
but the same kind of idea is there this
idea that
because obviously i know it's sleep but
it's still an intentional state
uh it wasn't and you'd expect someone to
always wake up
because of the loudness it is an
attentional
state absolutely and it's it's
encultured and it's developed
and somebody could have phases or you
know all these things can come into play
it is like
is the noise hitting the eardrum yes are
the neurons firing
that's where predictive processing has a
lot to say and that's why it was an
important theory because the centralized
processing theory
would be like it hits the baby's eardrum
it goes full

force loud sound incoming and then
there's the naptime
region and that just says nope i'm
asleep we just don't listen to sound
when i'm asleep
the predictive processing model is that
there's a generative
capacity like we're sleeping almost
semantically
and that propagates in a way that
dampens
the ignition of loud sounds so the net
effect is loud sounds don't wake the
sleeping baby
but it's extremely different if there's
like the central commander
that's receiving the raw sensory input
and processing it's not what happens
and it is much more like this generative
semantic capacity of systems to have
meaningful modes
that are distractible or not
distractible to different stimuli
so pretty nice good good baby story
so here's another phenomena that they're
going to
recapitulate so let's expand it in
6. so this is the inattentional
blindness task and it's just
another task where basically they're
going to differentiate a few
different uh experimental treatments
that usually would be used to pull out
attentional intentional blindness but
sort of just the summary of this is
in uh phase three which
uh was the self report you can read the
caption for

what they are but similarly depending on
which pieces and phases you add in
you get the existence of the simulated
erp
and a high fraction of having seen it if
there was attention
but then in the case where it wasn't
seen
then it just is like a totally flat
piece
so this is just uh then from the caption
says
in uh phase two firing rate plots
illustrate how more gradual updates in
the second level beliefs do not produce
p3 like erps
so it's interesting to think about how
they can actually titrate stimuli
and find out oh was the system updating
all of a sudden or was there more of
like continuous updating
but these are just kind of going into
specific um
examples that are like laid out in a
clear way but it's just it's a lot of
info that might be interesting for some
but
there's five more so let's just rapidly
kind of go through the figures yeah
figure seven is another layout of their
experimental results and again
it's a sort of spread where on the left
you have no
attention absence attention and the
right there is a regime of attention
parameter and then similarly weak
strength signal on top
strong strength signal on the bottom and

they're looking at the effect of a third
dimension
so this could have been like a hyper
cube of statistical results
and that's what this model and the
markov decision process allows
is instead of just making one model and
crafting it we could make like a
spectrum of 5000 models that are just
incrementally different and then report
the whole
distribution so here within each
quadrant
the left is inconsistent prior and the
right is a
consistent prior so here consistent
prior
strong signal high attention now we can
kind of imagine that's like
the focus you know paying only attention
to that and
blasts right in your face and it's like
100 saw it and 100 were correct
really salient stimuli really clear
self-report
no one's really doubting that they
perceived it with qualia these are the
easy cases
okay everything else is the gray zone
because if you have an inconsistent
prior again it's just a model
we're just experimenting and thinking
about it but it's interesting that
94 saw it 95 are correct
so even uh with high attention and high
signal strength
if your priors i'm not sure about what
i'm seeing

then 1 out of 20 people in the
simulation
again just loosely 1 out of 20 will
still not see it
but it was right there and they were
paying attention to it so there's a lot
of kind of
psychology anecdotes about stuff like
that like the whole room says that
line a is longer than line b and then
the last person just is like
i guess it's that way even though they
think that they're there to actually
speak the truth so
people can look at the percentages and
read more stories
but this is a very interesting modeling
framework because we can talk about how
these different
components of certainty and sense and
action are related in different
people's perspectives
i mean this this could be interesting as
well where it talks about this
um inconsistent prior
when you even have a strong um you know
a strong signal but if you've got an
inconsistent prior
you may not be aware um of something
and there's this this challenge in a way
where people
can be um tricked you know like we're
seeing with the internet and
cambridge analytics if you feed things
through
a level where it's not a strong enough
signal but it has associative
qualities you may be starting to

have that feed into your consciousness
at another level
but it's not activating your awareness
so i think that some of these questions
you know that
um you know and people might not notice
that happening because
when they do pay attention they you know
every they they they they think they've
seen everything there is to see but
there may be some things which are
just inconsistent with their priors as
much as they're weak
you know they could be and there's a
there's a little bit of an interplay
potentially
there that could be going on which we're
not aware of
nice and then also like even on the
other extreme if it's consistent prior
but
low attention and weak signal you still
sometimes report it and you still might
be correct even at a higher
uh amount of correctness and so
a lot of times um belief and faith
are as are as if they're to be ridiculed
and that's just
so far off from the way that everything
is
figure eight is uh again another
two by two with the tension and signal
strength and they're just really going
into
uh detail now overlaying this
prior consistency inconsistency question
and adding a neutral prior as well uh
and now overlaying those three so to be

almost like there was three cells here
but now overlaying those as three traces
of the erp
so what they're showing is that you get
sort of
the prior is like a fine tune it's like
a tweak
but it remains pretty much to be the
case
that you see the categorical difference
uh amongst each quadrant and
interestingly
the one that dips down gives like a
quasi p3
is the consistent prior strong signal
low attention
so that's like you're not really paying
attention but something loud happens
and you're sure that it's possible
that's actually the one that comes
closest
to actually inducing uh and we can look
at here percent c95
low attention high signal strength so
it's pretty interesting that
it's easier to have again you have to
calibrate it's real people
but it's easier to notice something when
it's a strong signal
even if you weren't paying attention
than it is
if you um are paying attention but it's
a weak signal
but maybe there's values for which it's
one way or the other but it really
separates those two things like is this
a subtle point in the text
or is this person just not paying

attention to the sentence
and those are sort of two questions and
you want them both to be
clear and so that you can align the
regime of attention but
that's why this is such a cool framework
okay
we're not going to go into too much
detail here but this
is a experimental paradigm laid out with
sort of classical psychologist
style of uh quote independent
manipulation of
expectations stimulus strength and
attention so they're able to use
repetition and variation in a specific
way
so that you can disentangle certain
attributes
from the sequence of behavior which is
something that we actually heard about
from adam saffron and colin de young
in the first streams of this year with
the way that you could look beyond the
personality surveys to the cybernetic
big five personality traits
so this is kind of like similar it's
revealing higher order
parametric values from
behavioral sequences that disentangle or
just like a gambling paradigm might be
high risk low risk high reward low
reward
so you're disentangling a few different
attributes the person's model
it's kind of like that but for visual
consciousness
access attention and then just to go

through the a1 and a2 supplements just
before we
kind of hit some closing thoughts so
a1 is their specification of the first
level generative model matrices
so let's definitely have them
talk about these but if you'll remember
from the model stream
it's all about specifying the matrices
and the calculations are all about
matrix multiplication
small or big or nested series of
matrices or
lists of lists of lists and so each of
these
outcomes at each level when we're
looking at figure
3 this kind of colorful connectivity
model each one of these
are basically reflected by a row or a
value
in a matrix two spaces 17
matrices seeing a theme with the
matrices and how
they can represent different things like
attention
or a word so then a certain combination
of rows is i didn't see a square
or i see a square but there's semantic
meaning
to the composition of a matrix and then
you can do
probabilistic inference using that
matrix
and then the second supplemental figure
is the seconds
level matrix structure it's like
you thought the movie was long wait till

you see the higher level matrix
and here it's similar except now we're
dealing with these hidden joint states
and that's why that was the second level
in the model where these higher level
states that also deal with
the behavior of the agent so
nice figures by them and they really
laid it all out
yet still it's kind of a lot to think
about so let's have them
walk through what they are but i guess
with that i'll just
kind of bring it to the ideas and
questions where we can go
and see if we just talk for as long as
you want just what
what do you think or what was like where
did we take that or who do you think we
find that
interesting or important
yeah i think um well
now i think that we've we will find that
a lot of people who are trying to um
understand how this modeling can be
applied
into other contexts and i think there's
there's probably going to be more and
more of a movement
for people to offer modeling um
approaches
more broadly so there's going to be
people who are going to want to make
models there's people are going to want
to work with people
doing models people want to look at what
the implications of the models are so i
suppose

that's something that this paper is
definitely from what we've been talking
about
um and so maybe
even asking um ryan and chris
to for a little bit there about you know
how would someone maybe engage
as they're working with those types of
matrices and
looking at the outputs to sort of
put that into a trans-disciplinary team
in context you know um that could be
quite interesting
um because this is this is all
um you know it has this potential really
but then
without without you know asking too much
of them but maybe
you know because um that that sort of
thing
and maybe just how if there's
different ways when you start to think
about the
a transformation matrix and
the the b matrix and the d matrix you
know just how
those things can be um you know are
there any
good practices for just helping keep
your
uh head straight with all of that as
people start to
you know because i know for ryan and
christopher it's second nature you know
they
they they can switch in between but it's
um
it does take a little bit of um

uh it takes a bit of tacit knowledge to
start to really
embody what's going on in there and how
that translates
um into this kind of common language
and so um but uh you know this paper
i think gives a sense of that but it's
still um
hopefully that's where this live stream
might be useful for people is as we go
through that
um especially when it got into some of
the later
model scenarios um yeah i got a little
bit
it was a little bit like a lot of
processing to try and work out what
exactly was the paradigm that's going on
um and again that might be because this
is specific to people working in
visual consciousness and some of these
types of you know
experiments so um so that would be you
know that that
type of broad question i think might be
quite interesting to uh and maybe even
it will yield
other model streams in the future if
they're even interested
um that build on that but otherwise i
think that
that covers it for me nice yep well
well said yeah thanks for kind of
bringing the specifics the model but
also
all the ways in which especially through
teamwork and communication we can be
working on it

and having an impact and that's really
an interesting area
i can see why you're interested in it
but
what a fun discussion it is always good
to read
the work of christopher and ryan it's
really just
informative at multiple levels so we'll
close
with that thanks again steven for
participating here
and we'll hope that people are
interested in coming on live for
18.1 or 0.2 or if it's after those dates
then just leaving comments and kind of
continuing the discussion
that way well that's it